

The Algorithm Terminates in Poly Time

Thm: Each iteration takes $O(|E|)$ time.

Thm: There are (at most) $O(|V|^2)$
iterations

Together, these imply that we can find
a min weight perf. matching in $O(|E| \cdot |V|^2)$.

Proof of Thm 1

Follows from fact that each iteration amounts to building an M -alt tree.

Stops when we have found:

a. Aug path (Case 1)

b. Blossom (Case 2)

c. Cannot grow tree, or shrink
a blossom (Case 3).

Proof of Thm 2 — 1/6

In each iteration, the alg either:

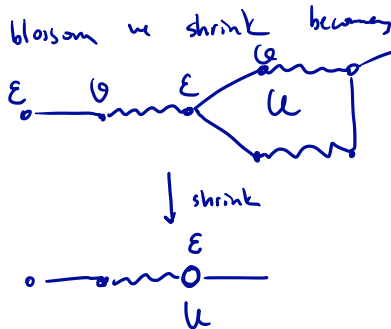
- ① • Augments M
- Shrinks a blossom and adds U to Ω .
- De shrinks an odd set U and removes U from Ω .

Clearly, M can only be augmented $O(|V|)$ times.

Proof of Thm 2 — 2/6

Lemma: A set U added to Ω (shrinking) cannot be removed (deshrinking) until after the next matching augmentation.

pf: Recall that any blossom we shrink becomes an even node:

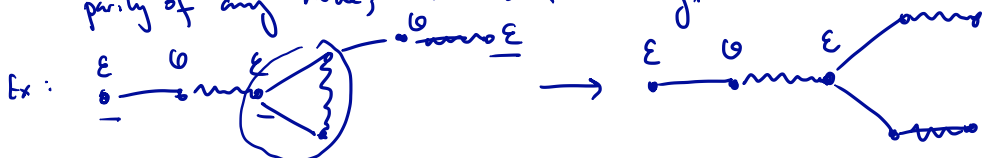


When we de-shrink because in

$y \leftarrow y + \theta d$, some y_u become zero.

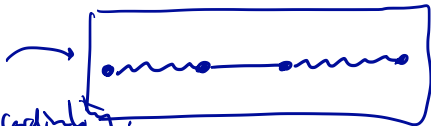
$$d_u = \begin{cases} +1 & \text{if } u \in E \\ -1 & \text{if } u \in \mathcal{O} \end{cases} \leftarrow \underline{\text{we only deshrink nodes in } \mathcal{O}.}$$

key: shrinking and deshrinking cannot change the parity of any node, nor can adding ^{new} edges



However:

The diagram shows a sequence of four elements: ϵ , 0 , ϵ , 0 . The first and third elements are connected by a solid line, and the second and fourth elements are connected by a wavy line. The first and third elements are highlighted with a pink background.



Only aug the matching can change cardinality.

Proof of Thm 2 — 4/6

What can happen b/w augmentations?

How many iterations?

By above Lemma:

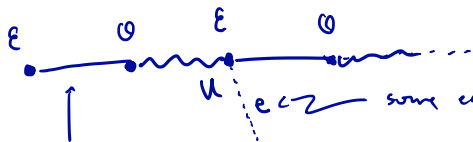
deshrinkings $\leq |\Omega|$, for Ω the laminar family after the last augmentation.

shrinkings $\leq |\Omega|$, Ω the laminar family before the next augmentation.

Ω laminar, $|\Omega| \leq O(n)$.

Proof of Thm 2 — 5/6

How many times can we add edges?



For edges already
in E' , these
remain tight

If one endpoint in E then
possible for: $\sum_{\substack{u: \\ e \in \delta(u)}} y_u \leq w_e$ could become tight.

We know: $w \notin O$ b/c then $\sum_{\substack{u: \\ e \in \delta(u)}} y_u$ does not change.

(i) $w \in E$: Once we reach

$y \leftarrow y + \theta d$, we now have added e to $E_y \Rightarrow w \in O$, and
therefore this can happen at most $|V|$ times (max # vertices of E)

(ii) $w \in E$: Then $w \in X$ and we have an $X-X$ alt. path in
next iteration.

Proof of Thm 2 — 6/6

Therefore, between augmentations:

$$\# \text{ deshrinkings} \leq |\Sigma^{\text{int}}| \leq 2|V|$$

$$\# \text{ shrinkings} \leq |\Sigma^{\text{fin}}| \leq 2|V|$$

$$\# \text{ added edges} \leq |V|$$

Hence: We have at most $O(|V|)$ augmentations,
and at most $O(|V|)$ iterations b/w augmentations.

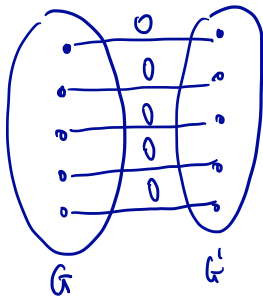
⇒ The alg. terminates in $O(|V|^2)$ iterations.

Finally: The alg. can only terminate with a perfect
matching in E' i.e. X empty ⇒ this can be expanded
to a perf. matching in E .

Max Weight Matching

First note that we can use the above
Primal-Dual alg. to find a max wt. perf
matching by flipping weights and
finding min wt. perf matching.

For a max wt matching (not nec perf).



Exercise: Verify that
a max wt perfect matching
in new graph, gives a max
wt matching in G .

Augmentation can change the parity

